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Bitcoin Sentiment & Trader Behavior Analysis · Primetrade.ai DS Task

What I Did

Data Merging

Merged the Bitcoin Fear & Greed Index (daily) with Hyperliquid trade records on date — each trade carries its corresponding sentiment label.

Exploratory Data Analysis

Studied PnL distribution, trade sizing, and win rates split across Fear vs. Greed regimes.

Feature Engineering

Built rolling account-level features on past trades only — win rate, avg leverage, trade frequency.

Trader Clustering

Applied KMeans clustering. Elbow plot suggested $k=4$. Clusters separate high-frequency accounts from large-size accounts.

Predictive Modeling

Trained a Random Forest classifier to predict Win / Loss. Detected and removed leaky feature 'Direction' which was correlated with the target.

Model Evaluation

Compared baseline accuracy (~ 0.58) vs model (~ 0.82). Produced confusion matrix and feature importance chart.

What I Found

Sentiment Affects Sizing, Not Outcomes Directly

- Greed regime traders take larger positions with wider size distribution.
- Fear regime shows conservative sizing — tighter clusters around smaller trades.
- Sentiment adds regime context but is not the primary driver of win/loss.

Top Predictive Features

- Fee and Size USD ranked #1 and #2 in feature importance.
- Coin (asset class) and Side (Long/Short) also in top 4.

4 Trader Archetypes Identified

- $k=4$ selected via elbow plot — best cluster separation.
- High-frequency small-account traders vs. low-frequency large-size traders clearly split.

PnL is Highly Skewed

- Most trades produce small losses; a few produce outsized gains.
- A small set of trades accounts for the bulk of positive PnL.

Feature Leakage Detected and Fixed

- 'Direction' was correlated with the target — removed before final modeling.
- After removal, accuracy settled at a realistic ~ 0.82 with balanced precision/recall.

Results

Baseline Accuracy

~ 0.58 (majority class)

Model Accuracy

~0.82 (Random Forest, leakage removed)

Precision / Recall

~0.79 / ~0.79 — balanced across Win and Loss

Confusion Matrix

Loss: 18,081 correct · 3,281 missed

Trader Clusters

k = 4 — chosen via elbow plot

Top Features

Fee · Size USD · Coin · Side

Leakage Removed

'Direction' — directly correlated with target label

Stack: Python 3.11 · pandas · scikit-learn · matplotlib · seaborn · Jupyter